The Optimal Fakeness of a Central Bank

Arbitrage, Fiscal Dominance, Credibility Capital, and Optimal Monetary Institutions in Peace and War

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Abstract

This paper develops a theory of the optimal institutional "fakeness" of a central bank. The term is used technically rather than pejoratively: fakeness measures the distance between the formal appearance of an independent central bank and its effective subordination to fiscal, political, strategic, or military objectives. In normal peacetime conditions, sufficiently competitive financial markets can penalize institutional fakeness through inflation expectations, currency substitution, capital flight, risk premia, parallel markets, and other forms of arbitrage. Under suitable conditions, the peacetime optimum is therefore complete institutional authenticity. War changes the objective function. Fiscal mobilization, emergency liquidity, strategic credit allocation, financial repression, capital controls, and coordination between monetary and fiscal authorities may generate positive returns to partial institutional subordination. Excessive subordination, however, destroys credibility, raises inflationary and exchange-rate risks, and can undermine the monetary capacity that wartime intervention is intended to increase. The resulting wartime optimum can therefore be interior.

The paper develops scalar and multidimensional models of central-bank fakeness, proves peacetime and wartime results, derives an optimal fakeness interval under Knightian uncertainty, treats credibility as a depreciable stock of institutional capital, formulates a stochastic dynamic program, derives comparative statics with respect to war intensity, studies postwar hysteresis and political-economy ratchets, and proposes an empirical state-space framework. Connections with finance, welfare economics, warfare economics, international relations, physics, chemistry, biology, psychology, philosophy, statistics, computer science, data science, and machine learning are used as disciplined structural analogies and modeling tools rather than as substitutes for economic identification. The central result is summarized by

$$F_P^* = 0, \qquad 0 < F_W^* < 1, \qquad F_{\text{postwar}}^* \to 0,$$

under the maintained assumptions of the benchmark model.

The paper ends with "The End"

Keywords: central banking, central-bank independence, fiscal dominance, war finance, monetary credibility, institutional economics, arbitrage, financial repression, dynamic programming, Knightian uncertainty, political economy.

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1 Introduction

Central banks are usually evaluated along dimensions such as independence, credibility, transparency, price stability, lender-of-last-resort capacity, and macroeconomic stabilization. These dimensions are not necessarily aligned in all states of the world. An institutional configuration that is desirable in peace may not be optimal during a large war, financial siege, existential emergency, or severe mobilization shock.

This paper studies that state dependence through a deliberately provocative but precisely defined concept: the *fakeness* of a central bank.

A fake central bank, in the terminology of this paper, is not a counterfeit bank, fraudulent organization, or fictitious institution. Rather, it is a legally recognizable central bank whose effective objective function, constraints, or operating procedures diverge from the conventional model of an independent monetary authority. The institution may preserve the outward architecture of central banking while becoming partially subordinated to fiscal, political, strategic, or military objectives.

Let

$$F \in [0, 1]$$

measure aggregate institutional fakeness. At one extreme,

$$F = 0$$

denotes complete institutional authenticity relative to the benchmark central-bank constitution. At the other,

$$F = 1$$

denotes complete effective subordination of the monetary authority.

The basic conjecture is state dependent.

In peace,

$$F_P^* = 0$$

may be optimal because markets penalize deviations from credible monetary institutions.

In war, however,

$$F_W^* > 0$$

may become optimal because the state obtains emergency benefits from monetary-fiscal coordination.

Yet complete subordination can destroy the credibility and monetary capacity required to finance the war itself. Hence

$$F_W^* < 1.$$

The benchmark prediction is therefore

$$\boxed{F_P^* = 0, \qquad 0 < F_W^* < 1, \qquad F_{\text{postwar}}^* \to 0.}$$

The paper develops this proposition formally and then extends it to multidimensional institutions, robust control, dynamic credibility, political economy, empirical estimation, and machine-learning-assisted policy analysis.

2 Conceptual Foundations

2.1 Authenticity and fakeness

Definition 2.1 (Central-bank authenticity). *Central-bank authenticity is the degree to which the effective behavior, objective function, constraints, and governance of a central bank correspond to its publicly stated institutional constitution.*

Definition 2.2 (Central-bank fakeness). *Central-bank fakeness is the degree of economically relevant divergence between formal central-bank architecture and effective control, objectives, or behavior.*

This definition deliberately separates institutional fakeness from legal illegitimacy. A central bank can be completely legal and yet possess positive F.

Likewise, emergency coordination between a treasury and central bank does not imply that the central bank is entirely fake. The concept is continuous.

2.2 Multidimensional fakeness

Let

$$\mathbf{f} = (f_L, f_O, f_F, f_B, f_R, f_C, f_S, f_T)^\top \in [0,1]^8,$$

where

f_L departure from legal independence;

f_O departure from operational independence;

f_F departure from fiscal-financing independence;

f_B departure from balance-sheet independence;

f_R departure from interest-rate independence;

f_C departure from conventional capital and exchange-market independence;

f_S departure from neutrality in strategic credit allocation;

f_T departure from conventional transparency and informational independence.

Define an aggregate index

$$F = \boldsymbol{\omega}^\top \mathbf{f},$$

where

$$\omega_i \geq 0$$

and

$$\sum_{i=1}^{8} \omega_i = 1.$$

Therefore,

$$F \in [0,1].$$

The scalar model is useful for proofs and intuition, while the vector model is more appropriate for empirical work.

3 The Peacetime Arbitrage Mechanism

Suppose peacetime welfare is

$$W_P(\mathbf{f}) = Y_P(\mathbf{f}) - C_P(\mathbf{f}) - A_P(\mathbf{f}) - \Pi_P(\mathbf{f}) - D_P(\mathbf{f}),$$

where Y_P is conventional stabilization output, C_P represents administrative and distortion costs, A_P represents arbitrage and portfolio-reallocation costs, Π_P represents inflationary and monetary-instability costs, and D_P represents credibility destruction.

The phrase "arbitraged away" should be interpreted broadly. Institutional inconsistency may be disciplined by

(i) capital outflows;

(ii) currency substitution;

(iii) changes in forward exchange rates;

(iv) sovereign and private risk premia;

(v) gold or commodity substitution;

(vi) parallel exchange markets;

(vii) shortened contract maturities;

(viii) indexation;

(ix) higher nominal yields;

(x) migration toward alternative stores of value.

Thus markets need not literally eliminate the central bank. They can instead reduce the effectiveness of the policies that institutional fakeness was intended to facilitate.

Assumption 3.1 (Peacetime marginal discipline)**.** *For every component f_i,*

$$\frac{\partial W_P}{\partial f_i} < 0$$

throughout the feasible region.

Theorem 3.2 (Peacetime Authenticity Theorem)**.** *Under peacetime marginal discipline, the unique welfare-maximizing institutional configuration is*

$$\mathbf{f}_P^* = \mathbf{0}.$$

Consequently,

$$F_P^* = 0.$$

Proof. Take any feasible $\mathbf{f} \neq \mathbf{0}$. Since every component of $\mathbf{f}$ is nonnegative and welfare is strictly decreasing in each f_i, reducing every positive component toward zero strictly increases welfare. Hence no point other than $\mathbf{0}$ can maximize W_P. Therefore

$$\mathbf{f}_P^* = \mathbf{0}.$$

Since

$$F = \boldsymbol{\omega}^\top \mathbf{f},$$

it follows that

$$F_P^* = 0.$$

□

4 War as a Regime Change

Let

$$z_t \in \{P, W\}$$

denote peace and war.

War introduces additional benefits from institutional flexibility. Define

$$G_W(\mathbf{f}) = G_F(\mathbf{f}) + G_M(\mathbf{f}) + G_S(\mathbf{f}) + G_L(\mathbf{f}),$$

where the terms respectively represent fiscal capacity, mobilization capacity, strategic financing, and emergency liquidity.

Wartime welfare is

$$W_W(\mathbf{f}) = W_P(\mathbf{f}) + G_W(\mathbf{f}).$$

Therefore,

$$\nabla W_W(\mathbf{f}) = \nabla W_P(\mathbf{f}) + \nabla G_W(\mathbf{f}).$$

At $\mathbf{f} = \mathbf{0}$ it is possible that

$$\nabla G_W(\mathbf{0}) > -\nabla W_P(\mathbf{0})$$

componentwise for at least some dimensions.

Then

$$\nabla W_W(\mathbf{0}) > 0$$

in those directions, and complete authenticity ceases to be optimal.

5 The Scalar Benchmark

Consider the reduced-form wartime objective

$$W_W(F) = W_0 + \alpha F - \beta F^2,$$

where

$$\alpha > 0, \qquad \beta > 0.$$

The parameter α captures the first-order strategic value of institutional flexibility, while β captures increasing credibility, inflation, arbitrage, and institutional costs.

The first-order condition is

$$\frac{dW_W}{dF} = \alpha - 2\beta F = 0.$$

Hence

$$F_W^* = \frac{\alpha}{2\beta}.$$

Moreover,

$$\frac{d^2 W_W}{dF^2} = -2\beta < 0.$$

Theorem 5.1 (Interior Fakeness Theorem)**.** *Suppose*

$$0 < \alpha < 2\beta.$$

Then wartime welfare has a unique interior maximum satisfying

$$0 < F_W^* < 1,$$

with

$$F_W^* = \frac{\alpha}{2\beta}.$$

Proof. Because $\alpha > 0$ and $\beta > 0$,

$$F_W^* = \frac{\alpha}{2\beta} > 0.$$

The condition $\alpha < 2\beta$ implies

$$\frac{\alpha}{2\beta} < 1.$$

Finally,

$$W_W''(F) = -2\beta < 0,$$

so W_W is strictly concave. Therefore the stationary point is the unique global maximum. □

The result formalizes the central intuition: too little wartime institutional flexibility can be costly, but too much destroys the monetary institution that generates the flexibility.

6 An Optimal Range Rather Than an Optimal Point

The policymaker may not know the true structural parameter vector θ.

Let

$$\theta \in \Theta,$$

where Θ is an ambiguity set.

Welfare becomes

$$W(F;\theta).$$

For tolerance $\varepsilon \geq 0$, define the robust near-optimal set

$$\mathcal{F}_\varepsilon = \left\{ F \in [0,1] : W(F;\theta) \geq \max_{x \in [0,1]} W(x;\theta) - \varepsilon \quad \text{for all } \theta \in \Theta \right\}.$$

If this set is nonempty and connected, write

$$\mathcal{F}_\varepsilon = [F_L, F_U].$$

Then the wartime institutional problem is not necessarily

$$F = F_W^*,$$

but rather

$$\boxed{F_L \leq F \leq F_U.}$$

The lower bound can be interpreted as a minimum emergency-flexibility constraint, while the upper bound is a credibility-preservation constraint.

7 War Intensity

Let

$$w \in [0,1]$$

denote war intensity.

Consider

$$W(F,w) = W_0 + \alpha(w)F - \beta F^2,$$

with

$$\alpha'(w) > 0.$$

The optimum is

$$F^*(w) = \frac{\alpha(w)}{2\beta}.$$

Therefore

$$\frac{dF^*}{dw} = \frac{\alpha'(w)}{2\beta} > 0.$$

Proposition 7.1 (War-Intensity Proposition)**.** *Conditional on an interior solution and constant β, greater war intensity increases the optimal degree of central-bank fakeness if the marginal strategic value of institutional flexibility rises with war intensity.*

One may interpret the regime continuum as

$$\text{peace} \rightarrow \text{mobilization} \rightarrow \text{limited war} \rightarrow \text{major war} \rightarrow \text{existential war}.$$

The theory therefore does not require a crude binary characterization of international conflict.

8 A General Comparative-Statics Model

Let

$$F^* = F^*(w, K, \pi^e, \sigma_A, \phi_F, \tau),$$

where

w war intensity;

K accumulated credibility capital;

π^e expected inflation;

σ_A intensity of arbitrage or speculative pressure;

ϕ_F emergency fiscal-financing requirement;

τ expected duration of the emergency.

A useful reduced form is

$$W = W_0 + (a_w w + a_\phi \phi_F - a_\pi \pi^e - a_A \sigma_A)\, F - \beta F^2.$$

Then

$$F^* = \frac{a_w w + a_\phi \phi_F - a_\pi \pi^e - a_A \sigma_A}{2\beta},$$

subject to projection onto $[0, 1]$.

Under positive coefficients,

$$\frac{\partial F^*}{\partial w} > 0,$$

$$\frac{\partial F^*}{\partial \phi_F} > 0,$$

$$\frac{\partial F^*}{\partial \pi^e} < 0,$$

and

$$\frac{\partial F^*}{\partial \sigma_A} < 0.$$

The effect of K can be ambiguous. A highly credible central bank has a larger credibility buffer, but precisely because its credibility is valuable, destroying it can have a larger opportunity cost.

9 Credibility as Institutional Capital

Let

$$K_t \geq 0$$

be the stock of monetary credibility.

Suppose

$$K_{t+1} = (1 - \delta F_t)K_t + \rho(1 - F_t),$$

where

$$\delta > 0, \qquad \rho > 0.$$

The first term captures depreciation of accumulated credibility when the institution is subordinated. The second represents credibility formation under authentic institutional behavior.

The marginal effect of fakeness is

$$\frac{\partial K_{t+1}}{\partial F_t} = -\delta K_t - \rho < 0.$$

Thus fakeness consumes credibility capital.

This creates an intertemporal resource-allocation problem: wartime governments may convert institutional credibility into current fiscal and strategic capacity.

10 Dynamic Programming

Let the state vector be

$$s_t = (K_t, z_t, w_t, \pi_t^e, \phi_{F,t}).$$

The government chooses

$$F_t \in [0, 1].$$

Its objective is

$$\max_{\{F_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \gamma^t u(s_t, F_t),$$

where

$$0 < \gamma < 1.$$

The Bellman equation is

$$V(s) = \max_{F \in [0,1]} \left\{ u(s, F) + \gamma \mathbb{E}[V(s') \mid s, F] \right\}.$$

Suppressing state variables other than K, an interior first-order condition is

$$u_F + \gamma \mathbb{E} \left[V_K(K', z') \frac{\partial K'}{\partial F} \right] = 0.$$

Define the credibility shadow price

$$\lambda_t = V_K(s_t).$$

Then

$$u_F - \gamma \mathbb{E}_t \left[\lambda_{t+1} (\delta K_t + \rho) \right] = 0.$$

Hence the marginal current benefit of fakeness must equal the expected discounted shadow cost of destroyed credibility.

Proposition 10.1 (Credibility Shadow-Price Proposition)**.** *If*

$$V_K > 0,$$

then destruction of credibility imposes a strictly positive dynamic opportunity cost on wartime institutional subordination.

Proof. Since

$$\delta K_t + \rho > 0$$

and

$$V_K > 0,$$

the term

$$\gamma \mathbb{E}_t [V_K (\delta K_t + \rho)]$$

is positive. Therefore a policymaker choosing F_t must compensate for a positive intertemporal credibility cost. □

11 Finance and Asset Pricing

Central-bank fakeness should affect asset prices because monetary credibility enters discount rates and expected nominal payoffs.

Let a nominal government bond satisfy

$$P_t = \mathbb{E}_t \left[M_{t,t+1} X_{t+1} \right],$$

where $M_{t,t+1}$ is a stochastic discount factor and X_{t+1} is the nominal payoff.

Suppose the required nominal yield can be decomposed approximately as

$$i_t = r_t^* + \pi_t^e + RP_t + CP_t(F_t),$$

where CP_t is an institutional credibility premium satisfying

$$CP_t'(F_t) > 0.$$

Similarly, a stylized exchange-rate relation may be written as

$$\mathbb{E}_t \Delta s_{t+1} = i_t - i_t^* + \chi(F_t),$$

with

$$\chi'(F_t) > 0$$

under conditions in which institutional fakeness increases expected currency depreciation.

The financial system therefore provides an endogenous feedback mechanism:

$$F \uparrow \Rightarrow \text{credibility risk} \uparrow \Rightarrow RP \uparrow \Rightarrow \text{financing cost} \uparrow.$$

This mechanism helps generate the upper boundary F_U.

12 Welfare Economics

A wartime government should not maximize military financing alone. Let total social welfare be

$$W_t = U(C_t) - D(\pi_t) - L(Y_t - \bar{Y}_t) + S_t - H_t,$$

where C_t is civilian consumption, $D(\pi_t)$ is inflation damage, $L(\cdot)$ is an output-gap loss, S_t is national-security benefit, and H_t is the broader human and institutional cost of war.

If fakeness increases current strategic financing but damages civilian welfare, then

$$\frac{\partial S_t}{\partial F_t} > 0$$

can coexist with

$$\frac{\partial U(C_t)}{\partial F_t} < 0$$

and

$$\frac{\partial D(\pi_t)}{\partial F_t} > 0.$$

The social optimum therefore differs from a narrow military-budget optimum.

Remark 12.1. *This distinction is essential. A policy can increase the state's capacity to mobilize resources without increasing aggregate social welfare.*

13 Warfare Economics

Let wartime resources satisfy

$$Y_t = C_t + I_t + G_t^M + G_t^N,$$

where G_t^M denotes military expenditure and G_t^N other government expenditure.

Emergency monetary accommodation may relax a short-run financing constraint

$$G_t^M \leq T_t + B_t + \Delta M_t + \Omega_t,$$

where T_t is taxation, B_t debt issuance, ΔM_t monetary financing, and Ω_t external assistance or other resources.

If

$$\frac{\partial \Delta M_t}{\partial F_t} > 0,$$

then partial subordination can relax the wartime constraint.

But the resulting inflation tax is approximately

$$\mathcal{T}_{\pi,t} \approx \pi_t \frac{M_{t-1}}{P_t}.$$

Excessive reliance on this mechanism can shrink real money balances, so the inflation-tax base itself becomes endogenous.

This creates a Laffer-type wartime monetary-financing curve: initial monetary accommodation may increase real resources available to the state, while excessive accommodation can reduce them.

14 Political Economy and the Wartime Ratchet

Let social welfare be

$$W^S(F),$$

while the incumbent government's objective is

$$W^G(F) = W^S(F) + R(F),$$

where $R(F)$ denotes the private political value of monetary discretion.

Suppose

$$R'(F) > 0.$$

Then, under regularity conditions,

$$F_G^* > F_S^*.$$

Proposition 14.1 (Political Over-Subordination Proposition). *If policymakers obtain private political benefits from discretionary control of the monetary authority, the politically chosen degree of fakeness can exceed the socially optimal degree.*

War therefore creates a constitutional commitment problem.

A government may have a legitimate reason to move from

$$F = 0$$

toward

$$F_W^* > 0,$$

while simultaneously possessing incentives to move beyond

$$F_U$$

or to retain emergency powers after the war ends.

Institutional countermeasures can include sunset clauses, explicit financing ceilings, maturity limits, legislative authorization, ex-post audits, transparency requirements, and predetermined normalization rules.

15 Postwar Hysteresis

Suppose that after war ends at time T,

$$\dot{F}_t = -\kappa F_t,$$

where

$$\kappa > 0.$$

Then

$$F_t = F_T e^{-\kappa(t-T)}.$$

Therefore,

$$\lim_{t\to\infty} F_t = 0.$$

However, for finite $t > T$,

$$F_t > 0$$

whenever $F_T > 0$.

Thus restoration of institutional authenticity is gradual rather than instantaneous.

This yields a monetary hysteresis mechanism:

$$\text{peace} \to \text{emergency} \to \text{subordination} \to \text{peace} \not\Rightarrow \text{instantaneous restoration}.$$

16 International Relations and Geopolitics

Central-bank credibility has an external strategic dimension. Foreign governments, investors, banks, suppliers, and allies observe the monetary regime.

Let external financing capacity be

$$X_t = X(K_t, F_t, \mathcal{A}_t, \mathcal{S}_t),$$

where $\mathcal{A}_t$ denotes alliance support and $\mathcal{S}_t$ denotes sanctions or financial restrictions.

A plausible sign structure is

$$X_K > 0,$$

while

$$X_F < 0$$

may hold when fakeness reduces foreign confidence.

But sanctions can also alter the marginal value of domestic monetary control:

$$\frac{\partial^2 W}{\partial F \partial \mathcal{S}}$$

need not have a fixed sign.

Financial isolation can increase the value of administrative control, while simultaneously making domestic credibility more important.

Thus geopolitics changes both the benefits and costs of F.

17 A Strategic Game Between the State and Markets

Consider two players:

1. the state, choosing F;
2. financial markets, choosing portfolio exposure q.

Let state welfare be

$$W_G(F, q),$$

and market payoff be

$$W_M(q, F).$$

A Nash equilibrium satisfies

$$F^* \in \arg\max_F W_G(F, q^*)$$

and

$$q^* \in \arg\max_q W_M(q, F^*).$$

If market exposure falls with fakeness,

$$\frac{\partial q^*}{\partial F} < 0,$$

then policymakers face an endogenous market-discipline constraint.

This provides a game-theoretic interpretation of the claim that institutional fakeness can be arbitraged away.

18 Physics: Potential Functions and Phase Transitions

Physics provides a useful mathematical analogy.

Define a potential

$$\Phi(F; w) = -W(F; w).$$

The equilibrium institution minimizes Φ:

$$F^* = \arg\min_F \Phi(F; w).$$

As w changes, the location of the minimum can change.

For example,

$$\Phi(F; w) = \beta F^2 - \alpha(w)F.$$

Then

$$F^*(w) = \frac{\alpha(w)}{2\beta}.$$

A richer nonconvex potential could generate multiple institutional regimes and abrupt transitions.

This resembles a phase transition mathematically, but the analogy should not be interpreted as claiming that political institutions literally obey thermodynamic laws.

19 Chemistry: Activation Barriers and Institutional Reactions

Institutional transformation can also be represented using the mathematics of activation barriers.

Let the transition rate from authentic to emergency institutions be

$$r_{P \to W} = A \exp\left(-\frac{E_a - \eta w}{\mathcal{T}}\right),$$

where E_a is an institutional activation barrier, w is emergency intensity, $\eta > 0$, and $\mathcal{T}$ is a generic volatility or pressure scale.

Greater war intensity lowers the effective barrier to institutional change.

The analogy is structural: constitutional constraints act like barriers that prevent ordinary shocks from producing large regime changes, while extreme shocks can make previously inaccessible institutional configurations feasible.

20 Biology: Adaptation and Fitness

Biology suggests an adaptation interpretation.

Let institutional fitness in environment z be

$$\mathcal{G}(F, z).$$

The optimal phenotype of the monetary institution is

$$F^*(z) = \arg\max_F \mathcal{G}(F, z).$$

If environmental conditions change from peace to war, the fitness-maximizing institutional configuration may also change.

However, specialization carries costs. An institution optimized for war may perform poorly in peace.

The relevant analogy is therefore not "survival of the strongest" but state-contingent adaptation subject to adjustment costs.

21 Psychology, Psychiatry, and Decision-Making Under Stress

Severe emergencies change decision environments.

Let perceived war intensity be

$$\widehat{w}_t = w_t + \varepsilon_t,$$

where

$$\mathbb{E}[\varepsilon_t] = b_t.$$

Under unbiased perception,

$$b_t = 0.$$

Under systematic threat overestimation,

$$b_t > 0.$$

If

$$F^* = F^*(\widehat{w}),$$

with

$$\frac{dF^*}{d\widehat{w}} > 0,$$

then threat overestimation generates excessive institutional subordination.

Likewise, underestimation can generate insufficient emergency adaptation.

The psychiatric vocabulary of stress and impaired judgment can be relevant to individual decision-makers, but institutional outcomes should not be reduced to psychiatric diagnoses. The economically relevant object is biased or noisy decision-making under extreme stress.

22 Philosophy: Authenticity, Appearance, and Function

The word "fake" raises a philosophical problem.

An institution may be authentic in at least three senses:

1. **legal authenticity**: it is legally constituted;
2. **procedural authenticity**: it follows its stated rules;
3. **functional authenticity**: it performs the functions associated with its institutional identity.

These need not coincide.

A legally authentic central bank can become functionally subordinated. Conversely, an institution may temporarily violate ordinary procedures to preserve the deeper monetary functions for which it exists.

The normative problem is therefore not simply whether an institution is "real" or "fake." It is whether the divergence between appearance, procedure, and function improves social welfare under the relevant state of the world.

23 Statistics and Econometric Identification

The theory requires empirical measurement of an unobserved or partly observed institutional variable.

Let latent fakeness satisfy

$$F_t^* = \mathbf{x}_t^\top \boldsymbol{\beta} + u_t,$$

where $\mathbf{x}_t$ contains wartime and macro-financial covariates.

Observed institutional indicators satisfy

$$\mathbf{y}_t = \boldsymbol{\Lambda} F_t^* + \boldsymbol{\varepsilon}_t.$$

A state equation may be

$$F_t^* = \rho_F F_{t-1}^* + \beta_w w_t + \beta_\phi \phi_{F,t} + \beta_\pi \pi_t^e + \beta_A \sigma_{A,t} + u_t.$$

This creates a state-space model.

Under Gaussian disturbances,

$$u_t \sim N(0, Q)$$

and

$$\boldsymbol{\varepsilon}_t \sim N(\mathbf{0}, R),$$

the latent index can be estimated with a Kalman filter.

The central empirical hypothesis is

$$H_0 : \beta_w = 0$$

against

$$H_1 : \beta_w > 0.$$

A second hypothesis concerns nonlinear welfare effects:

$$W_t = \gamma_0 + \gamma_1 F_t + \gamma_2 F_t^2 + \mathbf{z}_t^\top \boldsymbol{\gamma} + e_t,$$

with the theory predicting

$$\gamma_1 > 0, \qquad \gamma_2 < 0$$

within wartime observations under the benchmark interior model.

The implied optimum is

$$\widehat{F}^* = -\frac{\widehat{\gamma}_1}{2\widehat{\gamma}_2}.$$

24 Causal Identification

Correlation between war and central-bank subordination is insufficient for causal inference.

War is endogenous to political and economic conditions, while institutional changes can themselves affect conflict capacity.

Potential empirical designs include

1. event studies around externally dated war shocks;
2. difference-in-differences designs;
3. synthetic controls;
4. local projections;
5. instrumental-variable strategies where defensible instruments exist;
6. historical panel methods;
7. structural estimation of the dynamic model.

A stylized difference-in-differences specification is

$$F_{it} = \alpha_i + \lambda_t + \beta(\mathrm{War}_i \times \mathrm{Post}_t) + \mathbf{x}_{it}^\top \boldsymbol{\gamma} + \varepsilon_{it}.$$

The identifying assumptions must be evaluated rather than presumed.

25 Data Science and Construction of a Fakeness Index

Suppose country i at time t has observable features

$$\mathbf{x}_{it} = (x_{1,it}, \dots, x_{p,it})^\top.$$

Candidate features include

1. statutory central-bank independence;
2. frequency of direct government financing;
3. central-bank holdings of government debt;
4. interest-rate intervention;
5. capital controls;
6. exchange controls;
7. directed credit;
8. emergency lending;
9. changes in disclosure practices;
10. inflation outcomes;
11. exchange-rate premia;
12. turnover of central-bank leadership.

A transparent linear index is

$$\widehat{F}_{it} = \sum_{j=1}^{p} \omega_j \widetilde{x}_{j,it},$$

where normalized variables satisfy

$$\widetilde{x}_{j,it} \in [0, 1].$$

Alternatively, principal components or latent-variable methods can estimate weights from the data.

Transparency is particularly important because an opaque black-box index would reproduce the very informational problem the paper seeks to measure.

26 Machine Learning

Machine learning can estimate nonlinear mappings

$$\widehat{F}^* = g_\theta(\mathbf{s}),$$

where $\mathbf{s}$ is the macro-financial and geopolitical state.

The loss function may be

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(F_i - g_\theta(\mathbf{s}_i))^2 + \lambda\Omega(\theta).$$

Possible estimators include regularized regression, random forests, gradient-boosted trees, neural networks, and Bayesian models.

Prediction alone, however, does not identify welfare-optimal policy. A machine-learning model trained on historical decisions estimates what governments did, not necessarily what they should have done.

Hence the appropriate architecture separates

$$\text{prediction}$$

from

$$\text{structural welfare evaluation.}$$

27 Reinforcement Learning and Artificial Policy Agents

The dynamic problem can be represented as a Markov decision process.

Let

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma),$$

where

$\mathcal{S}$ is the state space;

$\mathcal{A}$ is the action space of institutional choices;

P is the transition kernel;

R is social welfare;

γ is the discount factor.

An action can be the vector

$$\mathbf{f}_t \in [0,1]^8.$$

The optimal action-value function satisfies

$$Q^*(s,a) = R(s,a) + \gamma\mathbb{E}\left[\max_{a'} Q^*(s',a') \mid s,a\right].$$

A policy is

$$\pi(a \mid s).$$

For institutional applications, unconstrained reward maximization is inadequate. Impose constitutional constraints

$$g_j(s,a) \le 0, \qquad j = 1,\ldots,m.$$

The resulting problem is a constrained Markov decision process.

This is important because an algorithm maximizing short-run wartime output could otherwise learn policies that destroy long-run monetary credibility or civilian welfare.

28 A Robust-Control Extension

Suppose the policymaker distrusts the transition model.

Let P belong to an ambiguity set $\mathcal{P}$.

The robust Bellman equation becomes

$$V(s) = \max_{a\in\mathcal{A}} \min_{P\in\mathcal{P}} \left\{R(s,a) + \gamma \mathbb{E}_P[V(s')]\right\}.$$

This is particularly appropriate in war, where model uncertainty, structural breaks, missing data, strategic deception, and rapidly changing constraints can make conventional probabilistic assumptions unreliable.

29 Computer Science and Algorithmic Solution

A discretized dynamic-programming algorithm can approximate the optimal policy.

Algorithm 1: Value Iteration for Optimal Central-Bank Fakeness

Input: state grid $\mathcal{S}$, action grid $\mathcal{A}$, transition probabilities $P(s'|s,a)$, reward $R(s,a)$, discount factor γ, tolerance ϵ.

Initialize: $V_0(s) = 0$ for every $s \in \mathcal{S}$.

Repeat:
For each $s \in \mathcal{S}$:
For each $a \in \mathcal{A}$ compute

$$Q_k(s,a) = R(s,a) + \gamma \sum_{s'\in\mathcal{S}} P(s'|s,a)V_k(s').$$

Set

$$V_{k+1}(s) = \max_{a\in\mathcal{A}} Q_k(s,a).$$

Until

$$\max_s |V_{k+1}(s) - V_k(s)| < \epsilon.$$

Output:

$$\pi^*(s) = \arg\max_a Q(s,a).$$

30 Multidimensional Quadratic Model

The vector formulation permits interaction between different forms of institutional subordination.

Let

$$W_W(\mathbf{f}) = W_0 + \mathbf{a}^\top \mathbf{f} - \frac{1}{2}\mathbf{f}^\top H \mathbf{f},$$

where H is symmetric positive definite.

Then

$$\nabla_{\mathbf{f}} W_W = \mathbf{a} - H\mathbf{f}.$$

The unconstrained optimum satisfies

$$H\mathbf{f}^* = \mathbf{a},$$

and therefore

$$\boxed{\mathbf{f}^* = H^{-1}\mathbf{a}.}$$

Theorem 30.1 (Multidimensional Interior Solution)**.** *If H is positive definite and*

$$H^{-1}\mathbf{a} \in (0,1)^8,$$

then

$$\mathbf{f}^* = H^{-1}\mathbf{a}$$

is the unique interior wartime optimum.

Proof. Positive definiteness of H implies

$$\nabla^2 W_W = -H$$

is negative definite. Hence W_W is strictly concave. The unique stationary point is therefore the unique global maximizer. The stated condition places that maximizer in the interior of the feasible cube. □

The off-diagonal terms of H represent institutional interactions. For example, fiscal subordination and transparency loss may jointly create credibility costs larger than the sum of their separate effects.

31 Summary of Theoretical Predictions

Table 1: Benchmark comparative statics

Variable	Predicted effect on F^*	Economic mechanism
War intensity w	Positive	Mobilization value
Fiscal need ϕ_F	Positive	Financing flexibility
Expected inflation π^e	Negative	Credibility constraint
Arbitrage pressure σ_A	Negative	Market discipline
Credibility K	Ambiguous	Buffer versus opportunity cost
Political rents	Positive	Discretionary-control incentive
Postwar time	Negative	Institutional normalization

Table 2: Regime-dependent institutional logic

Regime	Principal objective	Main benefit of F	Main cost of F
Peace	Credibility and stabilization	Usually limited	Inflation, arbitrage, credibility loss
Mobilization	Preparation and liquidity	Coordination and flexibility	Early credibility deterioration
Major war	Resource mobilization	Fiscal and strategic capacity	Inflation and market withdrawal
Existential war	State survival	Maximum emergency flexibility	Possible monetary breakdown
Postwar	Reconstruction	Limited residual flexibility	Persistence of emergency institutions

32 Vector Graphics

32.1 The inverted-U wartime relationship

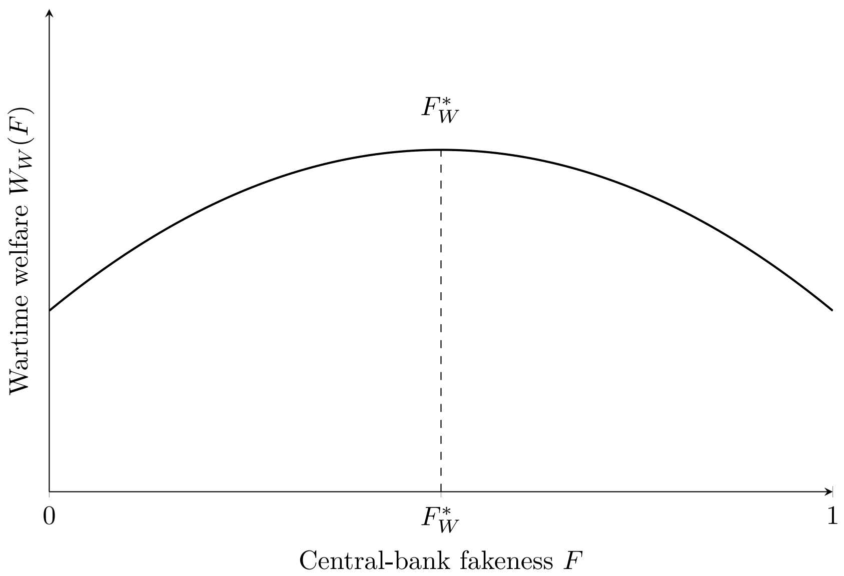

Figure 1: An interior wartime optimum. The figure is schematic rather than an empirical estimate.

32.2 Peace-war-postwar institutional dynamics

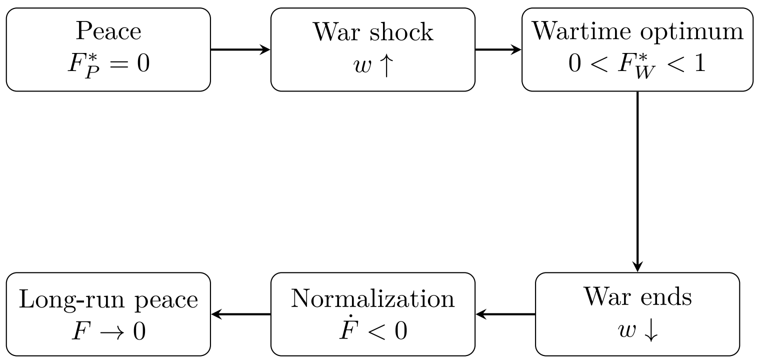

Figure 2: Regime-switching logic of optimal institutional fakeness.

32.3 Credibility-capital mechanism

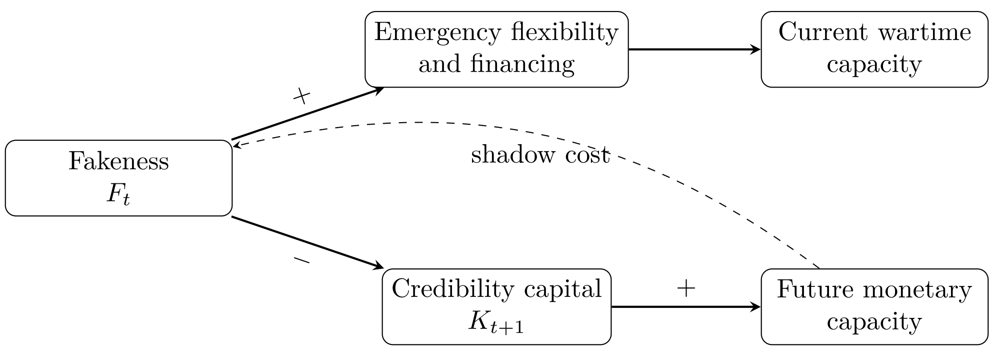

Figure 3: The intertemporal tradeoff between current flexibility and future credibility.

33 A Unified Systems Interpretation

The theory can be viewed as a feedback-control system.

Let

$$\mathbf{s}_{t+1} = A\mathbf{s}_t + B\mathbf{f}_t + G\boldsymbol{\varepsilon}_{t+1}.$$

Observed outcomes satisfy

$$\mathbf{y}_t = C\mathbf{s}_t + D\mathbf{f}_t.$$

The policymaker chooses $\mathbf{f}_t$ to stabilize a loss function

$$L_t = \mathbf{s}_t^\top Q\mathbf{s}_t + \mathbf{f}_t^\top R\mathbf{f}_t - \omega_S S_t.$$

In peace, R may place a high penalty on institutional departures.
In war, the weight on security S_t increases, altering the optimal feedback rule.
The policy can be written abstractly as

$$\mathbf{f}_t^* = \mathcal{K}(z_t)\mathbf{s}_t.$$

Hence the central-bank constitution itself becomes part of the state-contingent control architecture.

34 Limits of the Theory

Several qualifications are necessary.

First, the model does not establish that actual central banks should become less independent whenever war occurs. The result is conditional on the assumed benefits and costs.

Second, "fakeness" is a theoretical index. Empirical implementation requires careful operationalization.

Third, financial markets are not perfectly efficient. The peacetime arbitrage theorem is therefore a benchmark rather than a universal historical law.

Fourth, war is heterogeneous. A geographically remote limited conflict and an existential invasion need not imply similar monetary policies.

Fifth, emergency fiscal-monetary coordination can sometimes be achieved without large institutional fakeness if rules are explicit, temporary, credible, and transparent.

Sixth, causality is difficult to identify because monetary institutions, war, fiscal stress, inflation, and political regimes are jointly endogenous.

Seventh, cross-disciplinary analogies in this paper are mathematical or conceptual aids. Chemical kinetics, biological adaptation, and physical phase transitions are not offered as literal causal theories of central banking.

35 Policy Design

The theory suggests that the relevant institutional design problem is not simply

$$\text{independence versus dependence.}$$

It is

$$\text{state-contingent flexibility} + \text{credible bounds} + \text{automatic normalization.}$$

An emergency monetary constitution could therefore specify

1. the shocks that activate exceptional powers;
2. the dimensions of central-bank autonomy that may temporarily change;
3. quantitative financing ceilings;
4. maturity and collateral restrictions;
5. disclosure requirements;

6. legislative or judicial review;
7. periodic reauthorization;
8. sunset provisions;
9. postwar balance-sheet normalization;
10. procedures for rebuilding credibility capital.

The objective is to make

$$0 \to F_W^* \to 0$$

credible while preventing

$$0 \to F_W^* \to F_G^* > F_W^*$$

from becoming a permanent political ratchet.

36 Main Results

The theoretical architecture can be summarized as

$$\boxed{\begin{array}{c}\text{Financial arbitrage}\\ \downarrow \\ \text{Peacetime authenticity}\\ \downarrow \\ \text{War shock}\\ \downarrow \\ \text{Value of monetary flexibility}\\ \downarrow \\ \text{Interior institutional subordination}\\ \downarrow \\ \text{Credibility depletion}\\ \downarrow \\ \text{Postwar normalization.}\end{array}}$$

The principal benchmark result is

$$\boxed{F_P^* = 0, \qquad 0 < F_W^* < 1, \qquad F_{\text{postwar}}^* \to 0.}$$

Under ambiguity, the middle expression generalizes to

$$\boxed{F_W^* \in [F_L, F_U] \subset (0,1).}$$

Thus the central proposition is:

Authenticity dominates in peace; bounded institutional fakeness can dominate in war; restoration of authenticity dominates after war.

37 Conclusion

This paper has developed a state-contingent theory of central-bank institutional authenticity. In the benchmark peacetime environment, fakeness is disciplined by financial arbitrage, inflation expectations, portfolio substitution, credibility losses, and political-economy costs. Under the maintained monotonicity assumptions, complete authenticity is optimal.

War changes the objective function. Monetary-fiscal coordination can increase emergency liquidity, mobilization capacity, strategic credit, and fiscal flexibility. Consequently, the marginal value of some dimensions of institutional subordination can become positive.

The benefit is not unlimited. Excessive subordination can destabilize prices, weaken the currency, increase risk premia, accelerate currency substitution, reduce foreign financing capacity, and consume the stock of credibility on which future monetary effectiveness depends.

The resulting wartime problem is therefore naturally characterized by an interior optimum or, under ambiguity, an optimal interval:

$$0 < F_L \leq F_W^* \leq F_U < 1.$$

Treating credibility as capital produces a dynamic interpretation. Central-bank credibility accumulated during normal periods constitutes a strategic institutional reserve. Emergency governments can consume part of that reserve, but doing so has a shadow price because future governments inherit the resulting credibility stock.

The political-economy extension adds a second problem. Even when temporary subordination is socially optimal, incumbents may prefer greater or more persistent monetary discretion. Optimal wartime institutional design therefore requires both flexibility and a credible exit mechanism.

The theory ultimately replaces the binary question "Should the central bank be independent?" with a richer question:

Which dimensions of monetary independence should change, by how much, under which states of the world, for how long, and subject to which credibility-preserving constraints?

That formulation makes central-bank architecture itself an endogenous state-contingent economic variable.

References

[1] R. J. Barro and D. B. Gordon, "Rules, Discretion and Reputation in a Model of Monetary Policy," *Journal of Monetary Economics*, vol. 12, no. 1, pp. 101–121, 1983.

[2] B. S. Bernanke, *Essays on the Great Depression*, Princeton University Press, Princeton, 2000.

[3] M. D. Bordo and A. T. Levin, "Central Bank Digital Currency and the Future of Monetary Policy," NBER Working Paper No. 23711, 2017.

[4] P. Cagan, "The Monetary Dynamics of Hyperinflation," in M. Friedman, ed., *Studies in the Quantity Theory of Money*, University of Chicago Press, Chicago, 1956.

[5] G. A. Calvo, "Servicing the Public Debt: The Role of Expectations," *American Economic Review*, vol. 78, no. 4, pp. 647–661, 1988.

[6] A. Cukierman, *Central Bank Strategy, Credibility, and Independence: Theory and Evidence*, MIT Press, Cambridge, MA, 1992.

[7] M. Friedman, "The Role of Monetary Policy," *American Economic Review*, vol. 58, no. 1, pp. 1–17, 1968.

[8] F. E. Kydland and E. C. Prescott, "Rules Rather than Discretion: The Inconsistency of Optimal Plans," *Journal of Political Economy*, vol. 85, no. 3, pp. 473–491, 1977.

[9] R. E. Lucas, Jr., "Econometric Policy Evaluation: A Critique," in K. Brunner and A. H. Meltzer, eds., *The Phillips Curve and Labor Markets*, Carnegie-Rochester Conference Series on Public Policy, vol. 1, pp. 19–46, 1976.

[10] N. G. Mankiw, *Macroeconomics*, 10th ed., Worth Publishers, New York, 2019.

[11] C. M. Reinhart and K. S. Rogoff, *This Time Is Different: Eight Centuries of Financial Folly*, Princeton University Press, Princeton, 2009.

[12] T. J. Sargent, "The Ends of Four Big Inflations," in R. E. Hall, ed., *Inflation: Causes and Effects*, University of Chicago Press, Chicago, pp. 41–97, 1982.

[13] T. J. Sargent and N. Wallace, "Some Unpleasant Monetarist Arithmetic," *Federal Reserve Bank of Minneapolis Quarterly Review*, vol. 5, no. 3, pp. 1–17, 1981.

[14] C. A. Sims, "A Simple Model for Study of the Determination of the Price Level and the Interaction of Monetary and Fiscal Policy," *Economic Theory*, vol. 4, pp. 381–399, 1994.

[15] J. Tinbergen, *On the Theory of Economic Policy*, North-Holland, Amsterdam, 1952.

[16] C. E. Walsh, *Monetary Theory and Policy*, 4th ed., MIT Press, Cambridge, MA, 2017.

[17] M. Woodford, *Interest and Prices: Foundations of a Theory of Monetary Policy*, Princeton University Press, Princeton, 2003.

[18] R. Bellman, *Dynamic Programming*, Princeton University Press, Princeton, 1957.

[19] F. H. Knight, *Risk, Uncertainty and Profit*, Houghton Mifflin, Boston, 1921.

[20] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, Cambridge, MA, 2018.

[21] L. P. Hansen and T. J. Sargent, *Robustness*, Princeton University Press, Princeton, 2008.

[22] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[23] J. M. Keynes, *A Tract on Monetary Reform*, Macmillan, London, 1923.

[24] J. M. Keynes, *The General Theory of Employment, Interest and Money*, Macmillan, London, 1936.

[25] D. Ricardo, "Funding System," in *Supplement to the Fourth, Fifth and Sixth Editions of the Encyclopaedia Britannica*, Edinburgh, 1820.

Glossary

Arbitrage pressure
Portfolio and market responses that penalize inconsistent monetary institutions through prices, quantities, capital flows, or substitution.

Authenticity
Correspondence between a central bank's formal institutional constitution and its effective objectives, constraints, and behavior.

Central-bank fakeness
The economically relevant divergence between the formal appearance of central-bank independence and its effective institutional subordination. The term does not mean counterfeiting or fraud.

Credibility capital
A stock variable representing accumulated confidence in the monetary authority's ability and willingness to preserve the expected properties of money and monetary policy.

Credibility shadow price
The marginal continuation value of an additional unit of credibility capital.

Emergency monetary constitution
A system of state-contingent rules specifying how monetary institutions may temporarily change during an emergency.

Fiscal dominance
A condition in which fiscal requirements materially constrain or determine monetary policy.

Fakeness interval
The set $[F_L, F_U]$ of institutionally admissible wartime configurations that remain near-optimal or robust under uncertainty.

Financial repression
Policies that direct or constrain financial intermediation, asset holdings, interest rates, or capital movements in support of state objectives.

Hysteresis
Persistence of wartime institutional changes after the original shock has disappeared.

Institutional capital
: Accumulated organizational credibility, rules, conventions, and reputational assets that affect future policy effectiveness.

Knightian uncertainty
: Uncertainty for which the policymaker cannot confidently assign a single probability distribution.

Monetary authenticity
: The degree to which monetary institutions operate consistently with their declared constitution and policy framework.

Optimal fakeness
: The welfare-maximizing degree of central-bank institutional subordination.

Political ratchet
: The tendency for temporary emergency powers to persist because political actors obtain benefits from retaining them.

Robust control
: Optimization against a set of plausible models rather than a single known transition law.

War intensity
: A state variable measuring the economic and strategic severity of a military emergency.

Warfare economics
: The study of resource allocation, production, finance, mobilization, and welfare under military conflict.

The End